demo.py

```python

import cv2
from ultralytics import YOLO

def main():
    # Load the pretrained YOLO11 nano model
    model = YOLO("yolo11n.pt")

    # Open the default webcam
    camera = cv2.VideoCapture(0)

    if not camera.isOpened():
        print("Error: Could not open webcam.")
        print("Try changing VideoCapture(0) to VideoCapture(1).")
        return

    print("Real-Time Object Detection started.")
    print("Press 'q' to quit.")

    while True:
        success, frame = camera.read()

        if not success:
            print("Error: Could not read webcam frame.")
            break

        # Detect objects in the current frame
        results = model.predict(
            source=frame,
            conf=0.35,
            imgsz=640,
            verbose=False
        )

        # Draw bounding boxes, labels and confidence scores
        annotated_frame = results[0].plot()

        # Display the detection window
        cv2.imshow(
            "Real-Time Object Detection - YOLO11",
            annotated_frame
        )

        # Quit when q is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break

    camera.release()
    cv2.destroyAllWindows()
    print("Object detection stopped.")

```

```
52 if __name__ == "__main__":
53     main()
54
55
```